

App Dev Project Report

1. Student Details

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About Me:

I am a BS student with a strong interest in full-stack development and cloud technologies. I enjoy building scalable web applications and exploring real-world problem-solving through software system

2. Project Details

Project Title: Placement Portal Application (PPA)

Problem Statement:

The traditional placement process in colleges is often manual, unstructured, and inefficient. There is a need for a centralized system that allows students, companies, and administrators to seamlessly interact for managing placement drives and applications.

Approach:

The project was developed as a web-based application using **Flask** as the backend framework. The system follows a **role-based architecture**, where students, companies, and administrators have separate dashboards and functionalities.

Students can register, upload resumes, and apply for jobs. Companies can post placement drives and manage applications after admin approval. The admin has full control over approving companies, managing users, and monitoring placement activities.

The application uses **SQLAlchemy ORM** for database management and **Jinja2 templates with Bootstrap styling** for frontend rendering.

3. AI/LLM Declaration

I used **ChatGPT (GPT-5)** to assist in writing backend logic, debugging issues, and improving UI design.

The extent of AI/LLM usage is approximately **20–25%**, limited to:

- Code suggestions
- Debugging help
- Documentation formatting

All final implementation, integration, and testing were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for database interaction
Jinja2	Template engine for HTML rendering
Bootstrap 5	UI styling and responsive design
SQLite	Lightweight database
Werkzeug	Password hashing and security
HTML/CSS	Frontend structure and styling

5. Database Schema / ER Overview

Tables:

1. **Admin**
Stores admin credentials and manages the system.
2. **Student**
Stores student details such as name, email, course, skills, and resume path.
3. **Company**
Stores company details and approval status.
4. **PlacementDrive**
Contains job details like title, description, salary, and eligibility criteria.

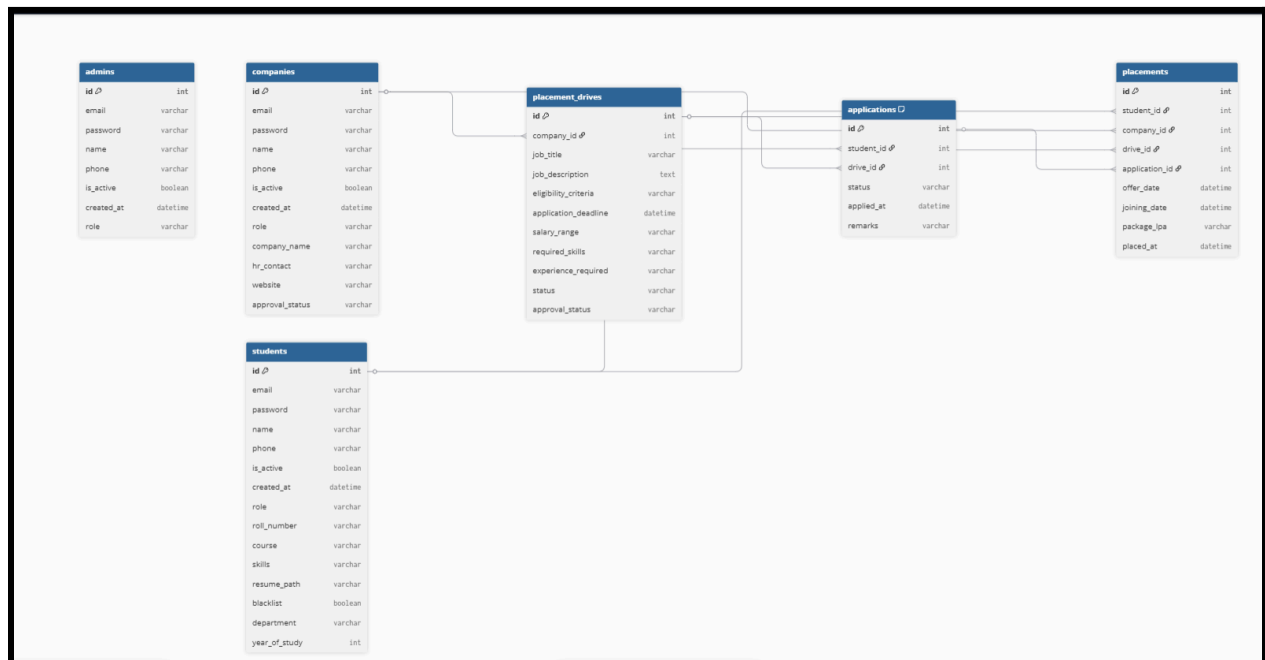
5. Application

Stores applications submitted by students for drives.

Relationships:

- One **Company** → **Many Placement Drives**
- One **Student** → **Many Applications**
- One **Placement Drive** → **Many Applications**

This forms a relational structure connecting students, companies, and job applications.



6. Route Endpoints

Endpoint	Method	Description
/login	POST	Authenticate user
/register/student	POST	Register new student
/register/company	POST	Register new company

<code>/dashboard/<role></code>	GET	Load dashboard based on role
<code>/create_drive</code>	POST	Create placement drive
<code>/apply/<drive_id></code>	POST	Apply to a job
<code>/admin</code>	GET	Admin dashboard

7. Architecture and Features

Architecture Overview:

- `app.py` → Main Flask application
- `models/` → Database models using SQLAlchemy
- `templates/` → HTML templates using Jinja2
- `static/` → CSS, images, and frontend assets

Implemented Features:

- Role-based authentication system
- Student registration and login
- Company registration with admin approval
- Placement drive creation and management
- Job application system
- Resume upload functionality
- Dashboard with statistics
- Admin control panel

Additional Features:

- Company approval and blacklist system
 - Application tracking
 - Session-based login handling
 - Clean UI with Bootstrap styling
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8. Video Presentation

Drive Link:

https://drive.google.com/file/d/1VBWPRGbNfwdzpwKvdAaCbS4_PwMTuWY0/view?usp=sharing